Digital Chip Design Program

Design RTL

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Task :05

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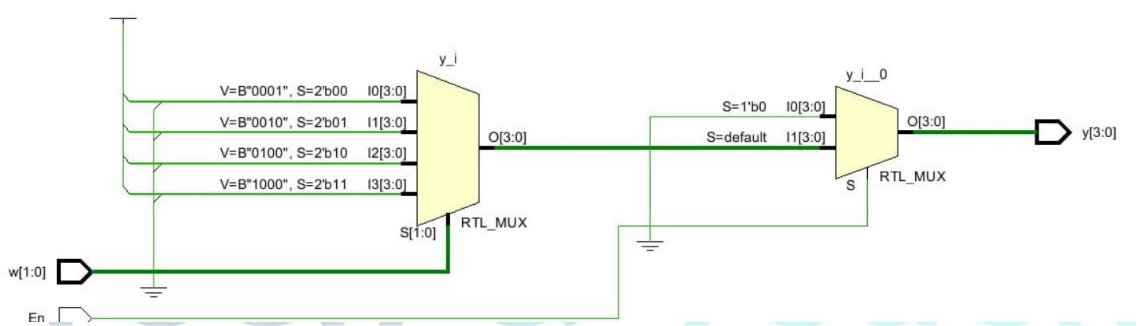
Task topic:

Combinational Datapath Design

Version 1.0

Task: 1.1.1

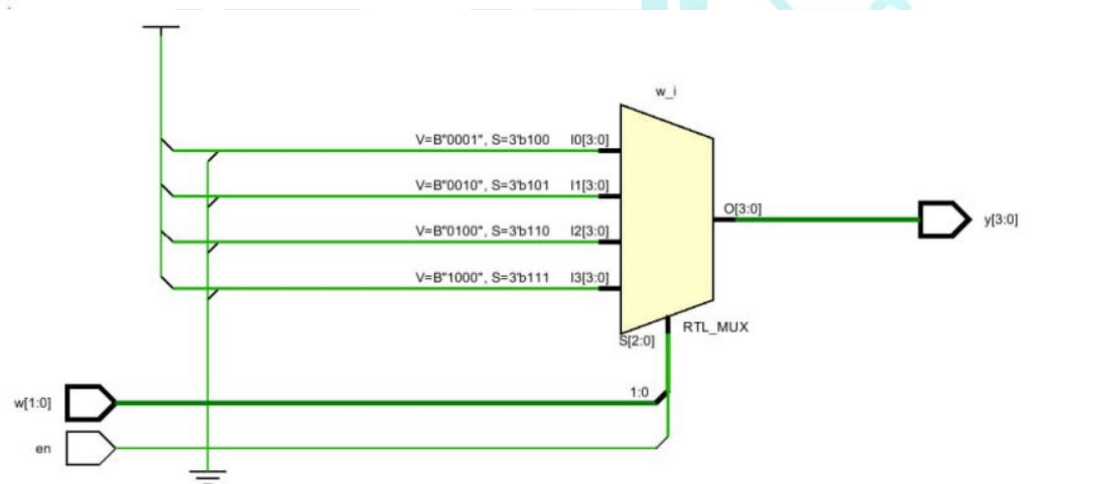
```
module Week2 (
    input              En,
    input        [1:0] w,
    output reg   [3:0] y
);

    always @(*) begin
        if (!En) begin
            y = 4'b0000;
        end
        else begin
            case (w)
                2'b00: y = 4'b0001;
                2'b01: y = 4'b0010;
                2'b10: y = 4'b0100;
                2'b11: y = 4'b1000;
                default: y = 4'b0000;
            endcase
        end
    end

endmodule
```



Task: 1.1.2

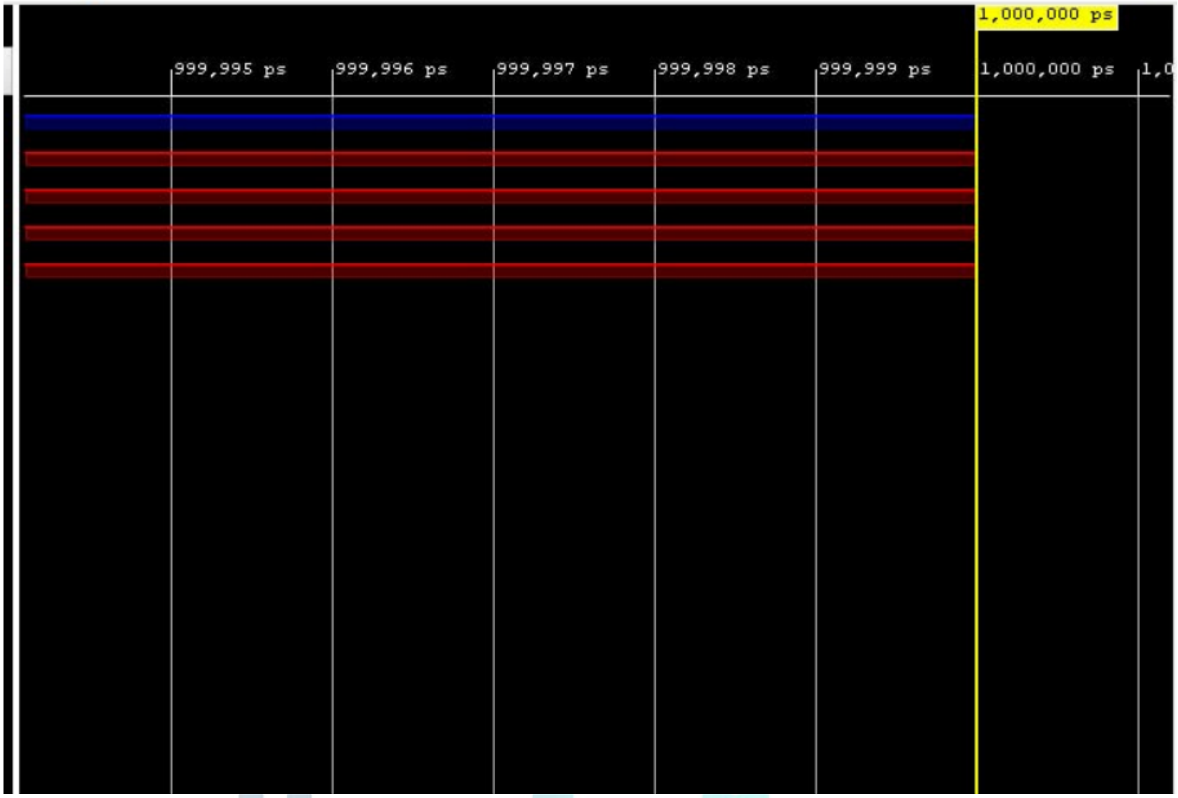
```
module decoder_1to2 (
    input  logic w,
    input  logic en,
    output logic [1:0] y
);
    assign y[0] = en & ~w;
    assign y[1] = en & w;
endmodule
```

```
module decoder_2to4 (
    input  logic [1:0] w,
    input  logic en,
    output logic [3:0] y
);
    assign y[0] = en & (w == 2'b00);
    assign y[1] = en & (w == 2'b01);
    assign y[2] = en & (w == 2'b10);
    assign y[3] = en & (w == 2'b11);
endmodule
```

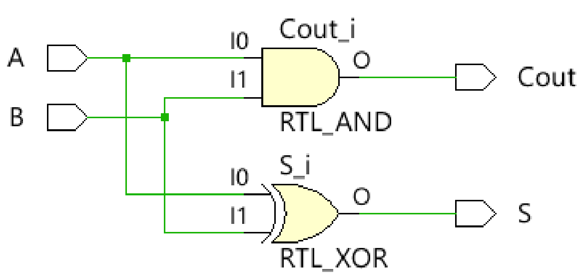


1.1.3

```
`timescale 1ns / 1ps
module Week2_Tasks1_1_3;
    logic [3:0] w;
    logic En;
    top_circuit uut (
        .w0(w[0]),
        .w1(w[1]),
        .w2(w[2]),
        .w3(w[3]),
        .En(En)
    );
    initial begin
        En = 0;
        w = 4'b0000;
        #10;
        En = 1;
        #10;
        w = 1;  #10;
        w = 2;  #10;
        w = 3;  #10;
        w = 4;  #10;
        w = 5;  #10;
        w = 6;  #10;
        w = 7;  #10;
        w = 8;  #10;
        w = 9;  #10;
    end
endmodule
```

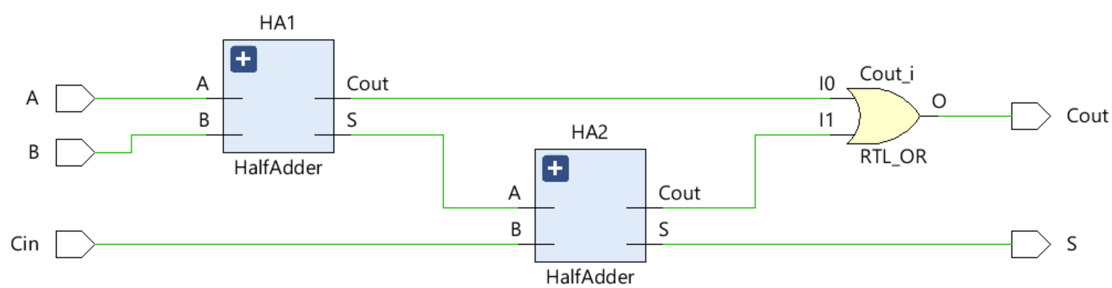

1,000,000 ps
999,995 ps
999,996 ps
999,997 ps
999,998 ps
999,999 ps
1,000,000 ps

Task: 1.2

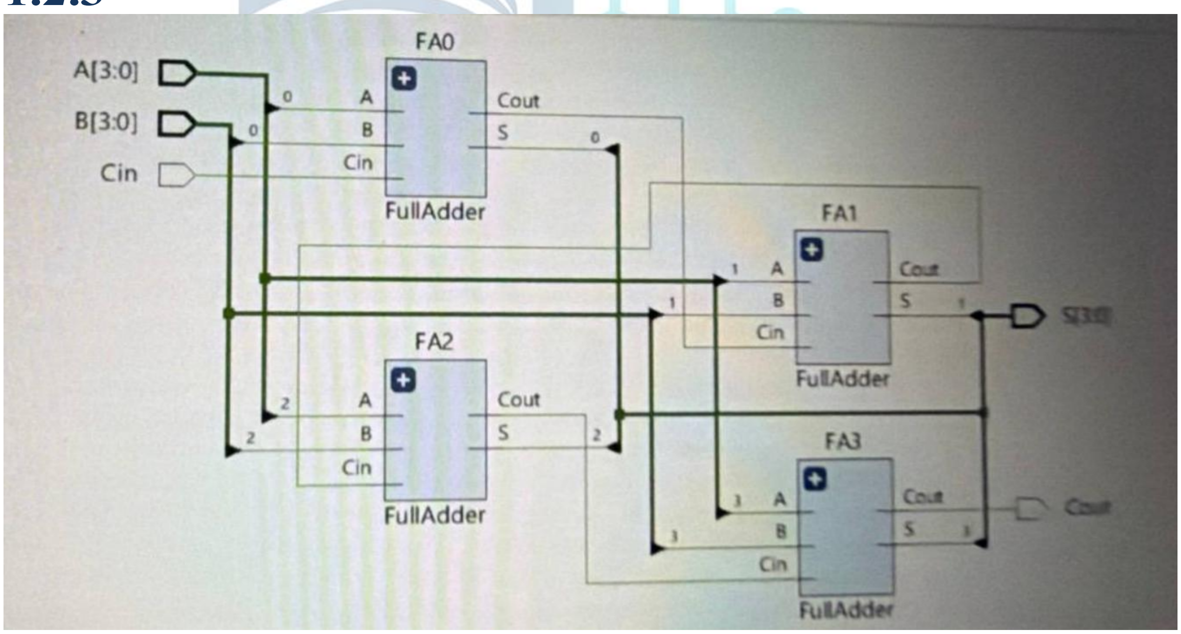

A
B
I0
I1
Cout_i
O
RTL_AND
Cout
I0
I1
S_i
O
RTL_XOR
S

1.2.2

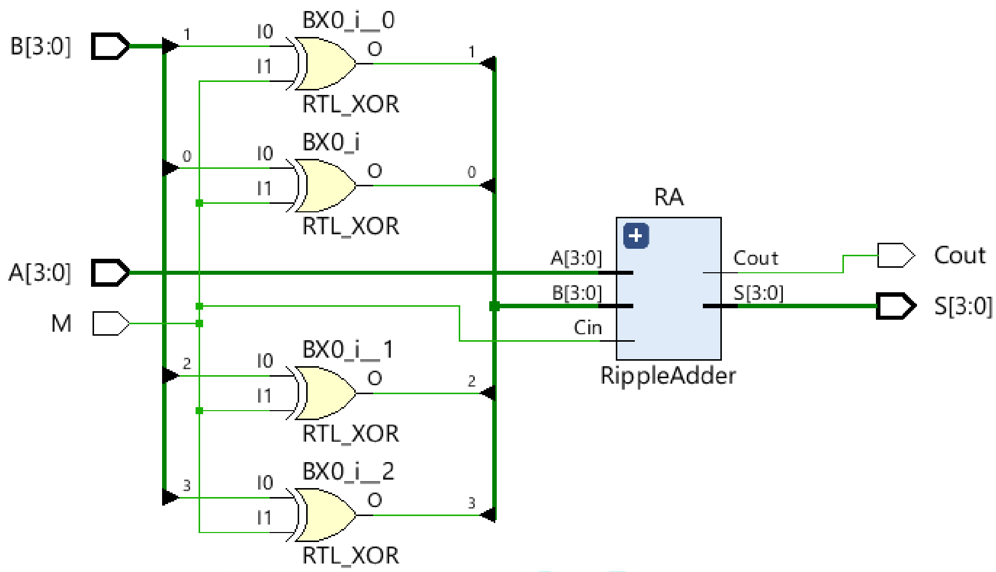
```
module HalfAdder(
    input logic A,
    input logic B,
    output logic S,
    output logic Cout
);

assign S = A ^ B;
assign Cout = A & B;

endmodule
```



1.2.3

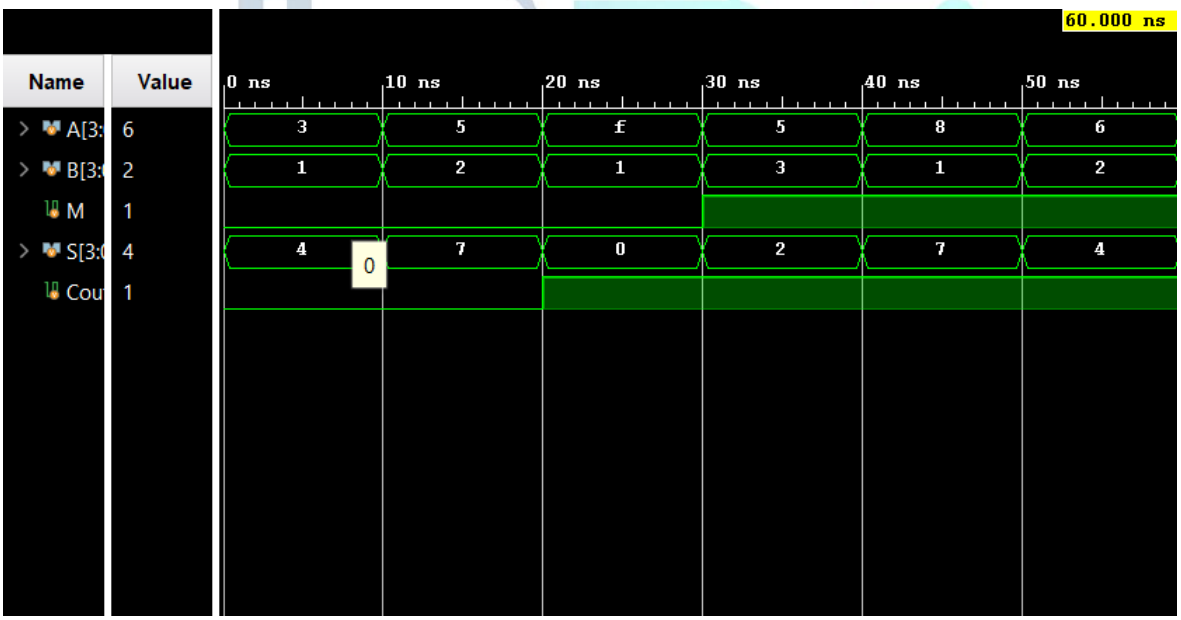


1.2.4



B[3:0]
A[3:0]
M
BX0_i__0
BX0_i
BX0_i__1
BX0_i__2
I0
I1
O
RTL_XOR
RA
A[3:0]
B[3:0]
Cin
Cout
S[3:0]
RippleAdder
Cout
S[3:0]

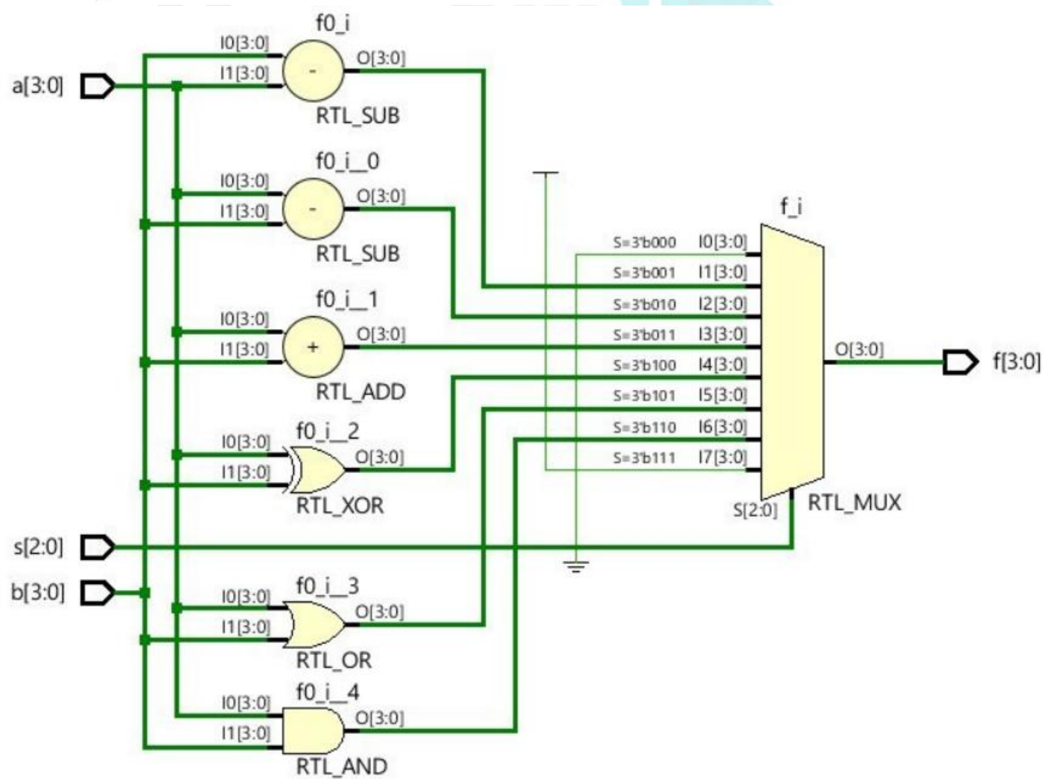
1.2.5



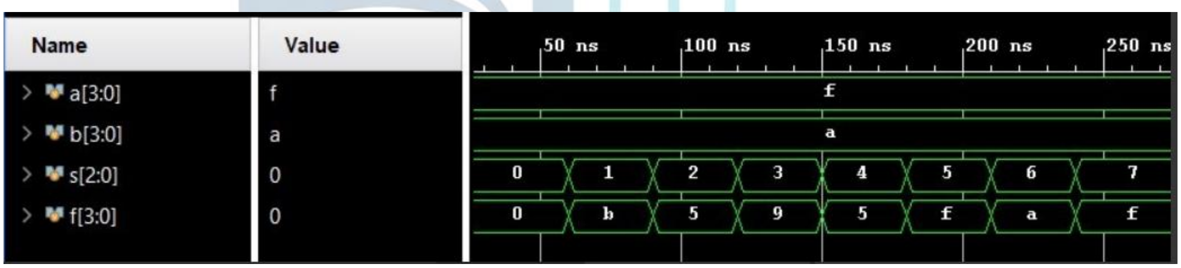
60.000 ns
Name
Value
A[3:
6
B[3:
2
M
1
S[3:
4
Cou
1
0 ns
10 ns
20 ns
30 ns
40 ns
50 ns
3
5
f
5
8
6
1
2
1
3
1
2
4
0
7
0
2
7
4

1.3

```
module ALU(
input logic [3:0] a,b,
input logic [2:0]s,
output logic [3:0]f
    );

    always_comb begin
    case(s)
    0 : f = 4'b0000;
    1 :  f = b - a;
    2 :  f = a - b;
    3 :  f = a + b;
    4 :  f = a ^ b;
    5 :  f = a | b;
    6 :  f = a & b;
    7 : f = 4'b1111;
    default : f = 4'bxxxx;
    endcase
    end
endmodule
```



```
`timescale 1ns / 1ps

module ALU_tb();
logic [3:0] a,b;
logic [2:0] s;
logic [3:0]f;


ALU dut(
.a(a),
.b(b),
.s(s),
.f(f)
);

initial begin
a = 4'b1111;
b = 4'b1010;
s = 0 ;

for(int i = 0 ; i < 8 ; i++)
  #30 s = i;
end
endmodule
```



Work distribution

Section	shaher	Suliman	Abdullah
Report	✓		
Task 1.1			✓
Task 1.2	✓		
Task 1.3		✓	